

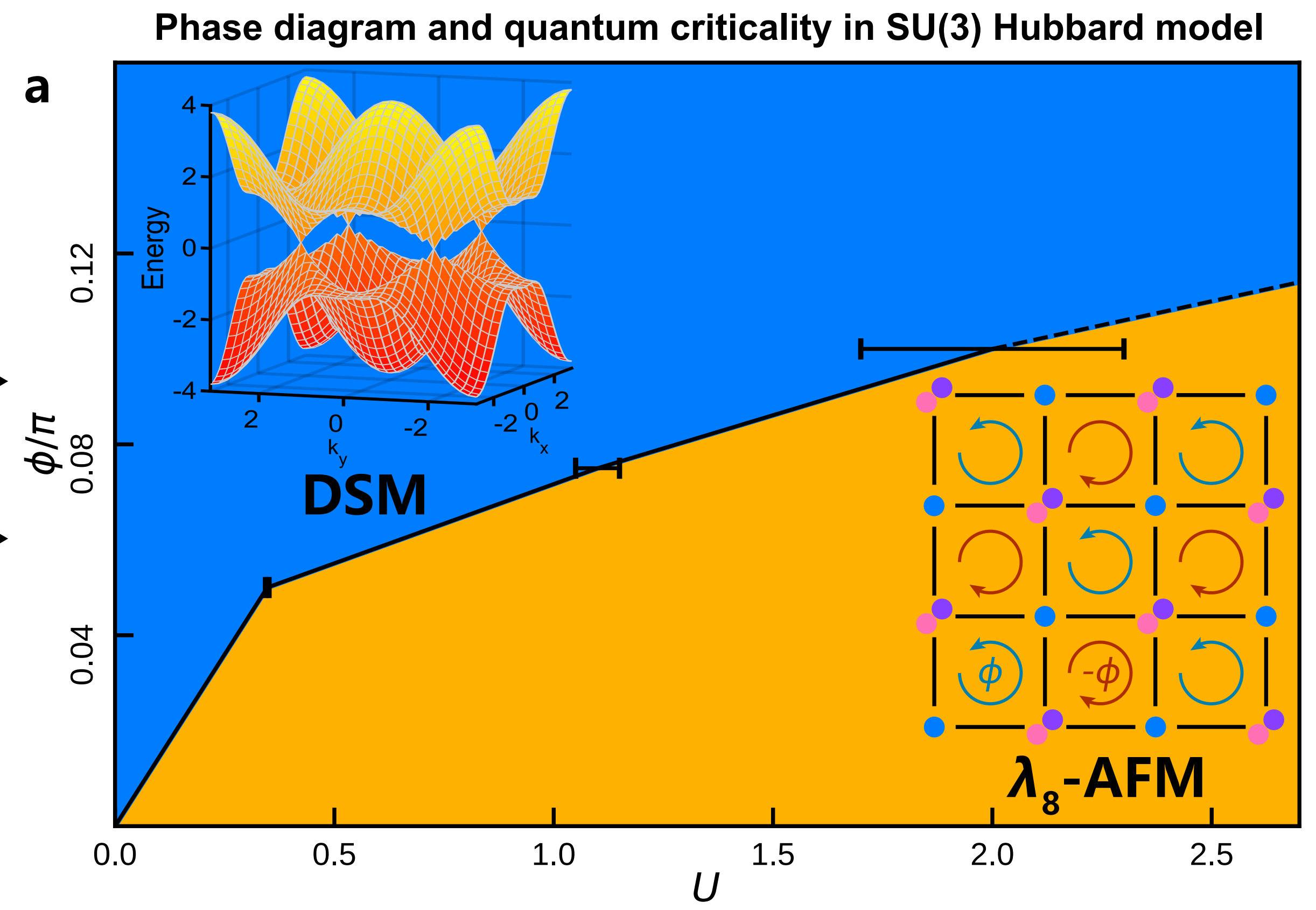
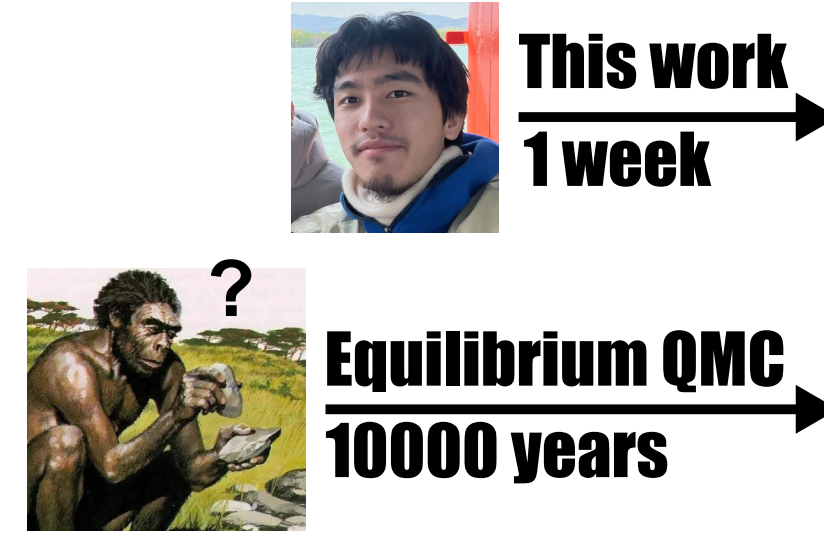
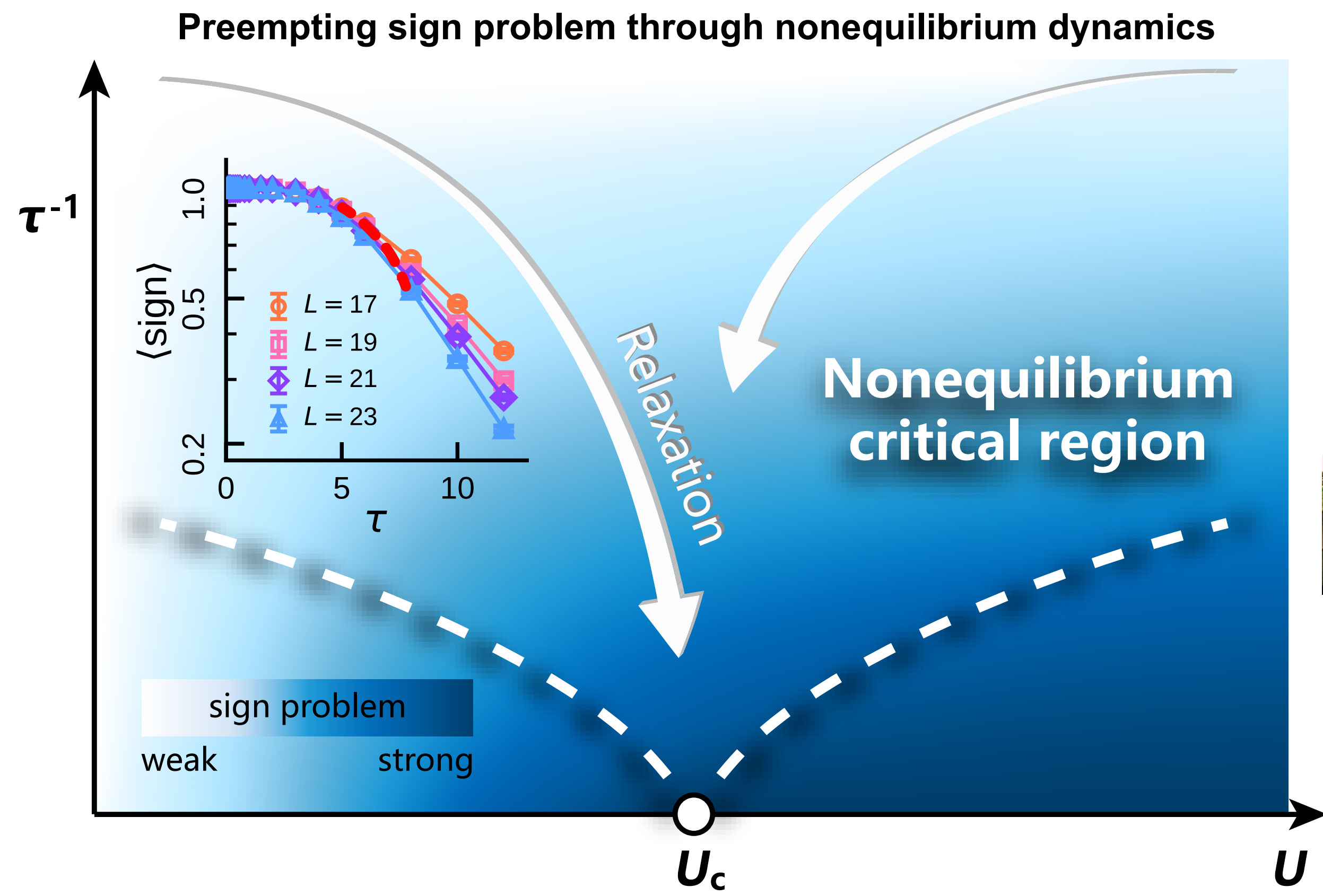
The 13th Workshop on Quantum Many-Body Computation, Hangzhou

Solving SU(3) Hubbard Model with Sign Problem: a Nonequilibrium Quantum Monte-Carlo Study

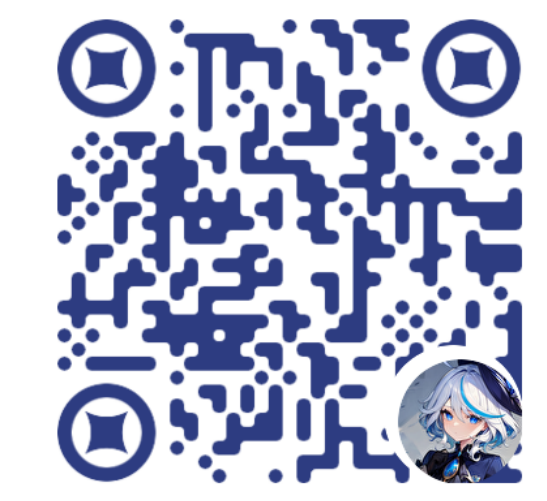
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Take-home concepts

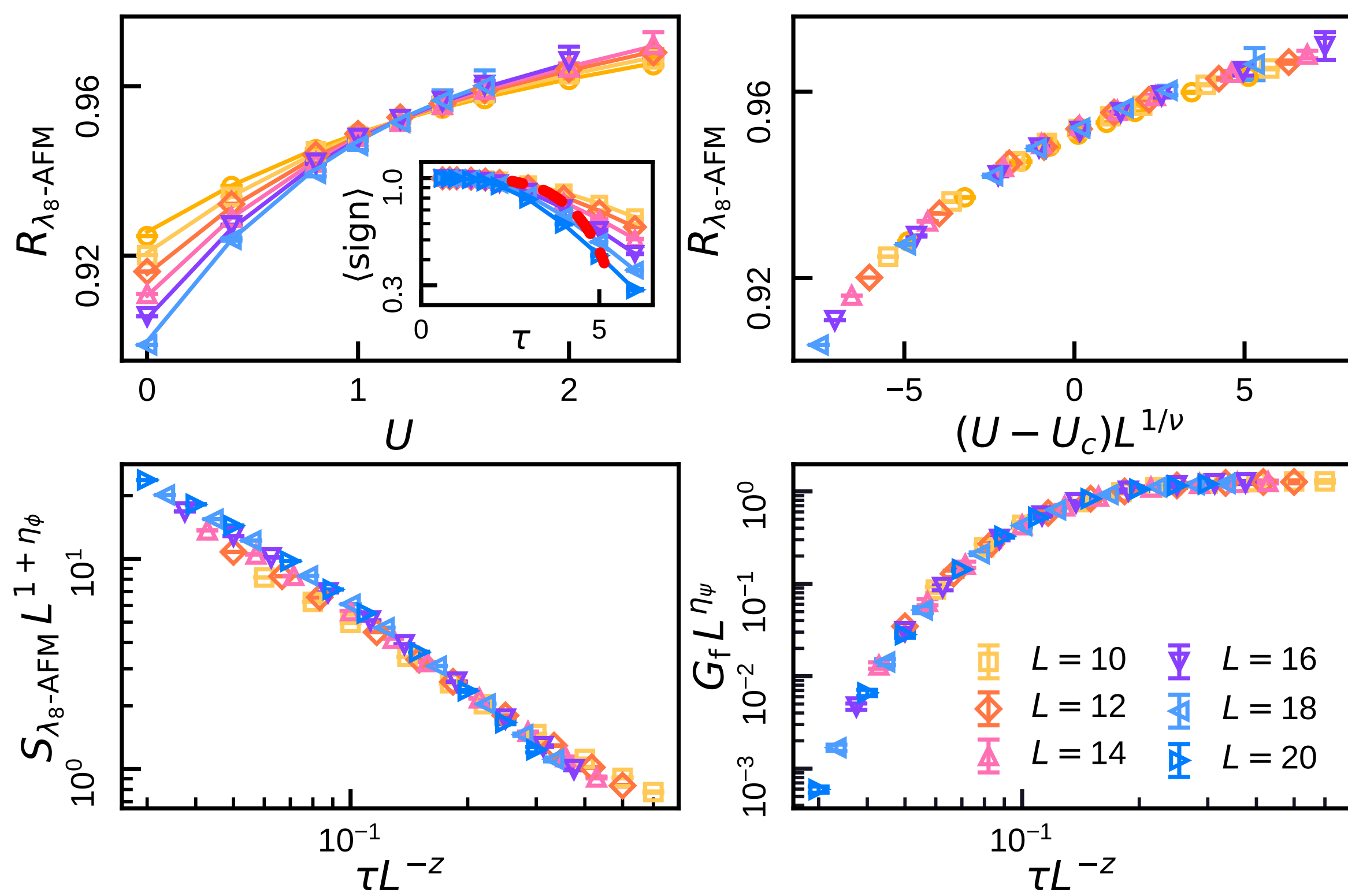


- New computational framework: nonequilibrium QMC**
Extract quantum critical point from nonequilibrium dynamics, circumventing severe sign problem.
- New phase of matter: λ_8 -AFM phase in SU(3) Hubbard model**
First discovery of λ_8 -AFM phase and its QCP with Dirac semi-metal in SU(3) Hubbard model.
- New universality class: chiral $\frac{SU(3) \times Z_2}{SU(2) \times U(1)}$ universality class**
Symmetry-breaking $SU(3) \times Z_2 \rightarrow SU(2) \times U(1)$, defining a novel universality class beyond known Gross-Neveu transitions.



arXiv:2410.18854

Quantum criticality



SU(3) Hubbard model with staggered flux

$$H = - \sum_{\langle ij \rangle, \alpha} t_{ij} c_{i\alpha}^\dagger c_{j\alpha} + U \sum_i \sum_{\alpha > \beta} (n_{i\alpha} - 1/2)(n_{i\beta} - 1/2)$$

where $U > 0$, $t_{ij} = e^{i\theta_{ij}}$. Magnetic flux in each plaquette is $\sum_{\square} \theta_{ij} = (-1)^{x+y}\phi$.

Observables and nonequilibrium scaling forms

- Order parameter: $m_{\lambda_8\text{-AFM}} \equiv \frac{1}{L^2} \sum_i (-1)^i \langle c_i^\dagger \lambda_8 c_i \rangle$
- Structure factor ($g = U - U_c$):
 $S_{\lambda_8\text{-AFM}}(\mathbf{k}) = \frac{1}{L^{2d}} \sum_{i,j} e^{i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)} (-1)^{i+j} \langle c_i^\dagger \lambda_8 c_i c_j^\dagger \lambda_8 c_j \rangle \rightarrow L^{-1-\eta_\phi} f_S(gL^{1/\nu}, \tau L^{-z})$
- Correlation-length ratio: $R_{\lambda_8\text{-AFM}} \equiv 1 - \frac{S_{\lambda_8\text{-AFM}}(\mathbf{k}=\Delta\mathbf{k})}{S_{\lambda_8\text{-AFM}}(\mathbf{k}=0)} \rightarrow f_R(gL^{1/\nu}, \tau L^{-z})$
- Fermion correlation: $G_f(\mathbf{k}) \equiv \frac{1}{L^2} \sum_{ij} e^{i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)} \langle c_{i,A}^\dagger c_{j,B} \rangle \rightarrow L^{-\eta_\psi} f_G(gL^{1/\nu}, \tau L^{-z})$

Critical properties probed under different staggered flux ϕ

ϕ	U_c	ν^{-1}	η_ϕ	η_ψ
0.05π	0.347(4)	0.67(1)	0.51(3)	0.16(2)
0.075π	1.10(5)	0.68(5)	0.55(5)	0.15(3)
0.1π	2.0(3)	0.58(7)	0.58(10)	0.16(3)

Symmetry breaking

SU(3) algebra

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda_8 = \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & -\frac{2}{\sqrt{3}} \end{pmatrix}$$

Only full-rank λ_8 can fully open spectral gap in Dirac fermions.

Four Goldstone modes: $\lambda_4, \lambda_5, \lambda_6, \lambda_7 \leftarrow [\lambda_a, \lambda_b] = 2i \sum_c f_{abc} \lambda_c$, $f_{458} = f_{678} = \frac{\sqrt{3}}{2}$.

Different Gross-Neveu-Yukawa universality classes in $d = 2 + 1$ with $N_f = 6$

Universality class	ν^{-1}	η_ϕ	η_ψ
chiral $\frac{SU(3) \times Z_2}{SU(2) \times U(1)}$ (this work)	0.68(5)	0.55(5)	0.15(3)
chiral Heisenberg ($4 - \epsilon$, 2nd order)[1]	1.478	1.023	0.058
chiral XY ($4 - \epsilon$, 2nd order)[1]	1.809	0.698	0.082
chiral Ising ($4 - \epsilon$, 2nd order)[1]	0.750	0.865	0.011
chiral Ising (FRG)[2]	0.993	0.912	0.013

Short-time nonequilibrium PQMC

Technical key points

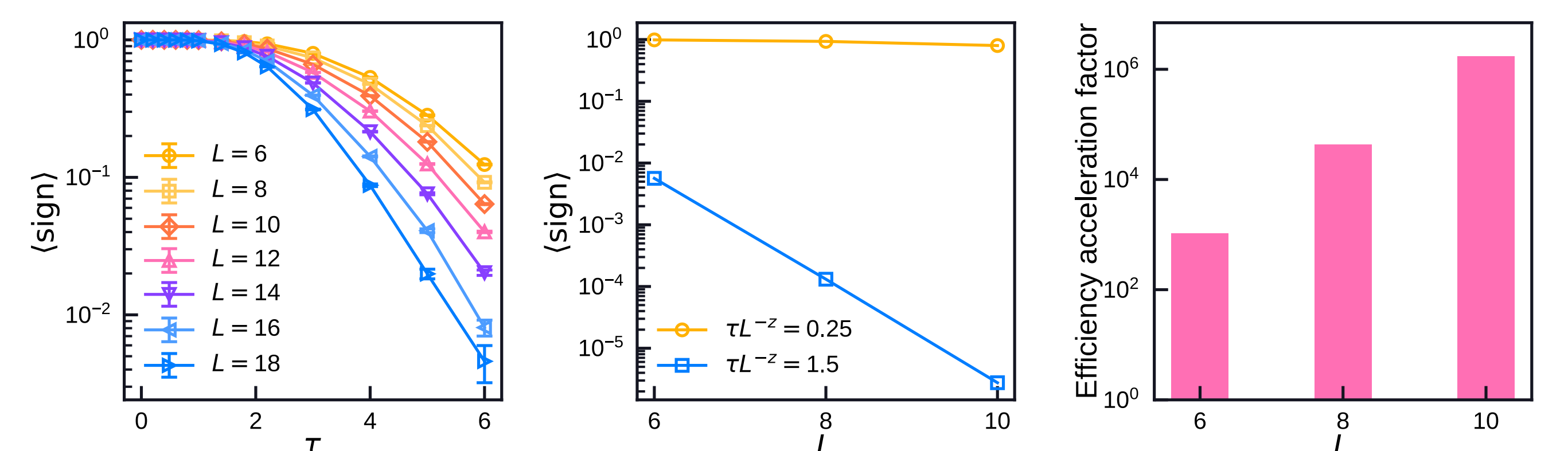
Imaginary-time relaxation from DSM/ λ_8 -AFM initial state [3, 4]:

$$\langle O(\tau) \rangle = \frac{\langle \psi_0 | e^{-\frac{\tau}{2}H} O e^{-\frac{\tau}{2}H} | \psi_0 \rangle}{\langle \psi_0 | e^{-\tau H} | \psi_0 \rangle}.$$

Set $\tau = 0.25L^z$ in this work, while typically $\tau = 2L^z \sim 3L^z$ for nonequilibrium.

Sign problem behaviors and computational efficiency

Very weak sign problem at short-time stage $\leftarrow \Delta \langle O \rangle \propto \frac{1}{\langle \text{sign} \rangle} \propto e^{\tau N \Delta f}$ [5]



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Reference

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